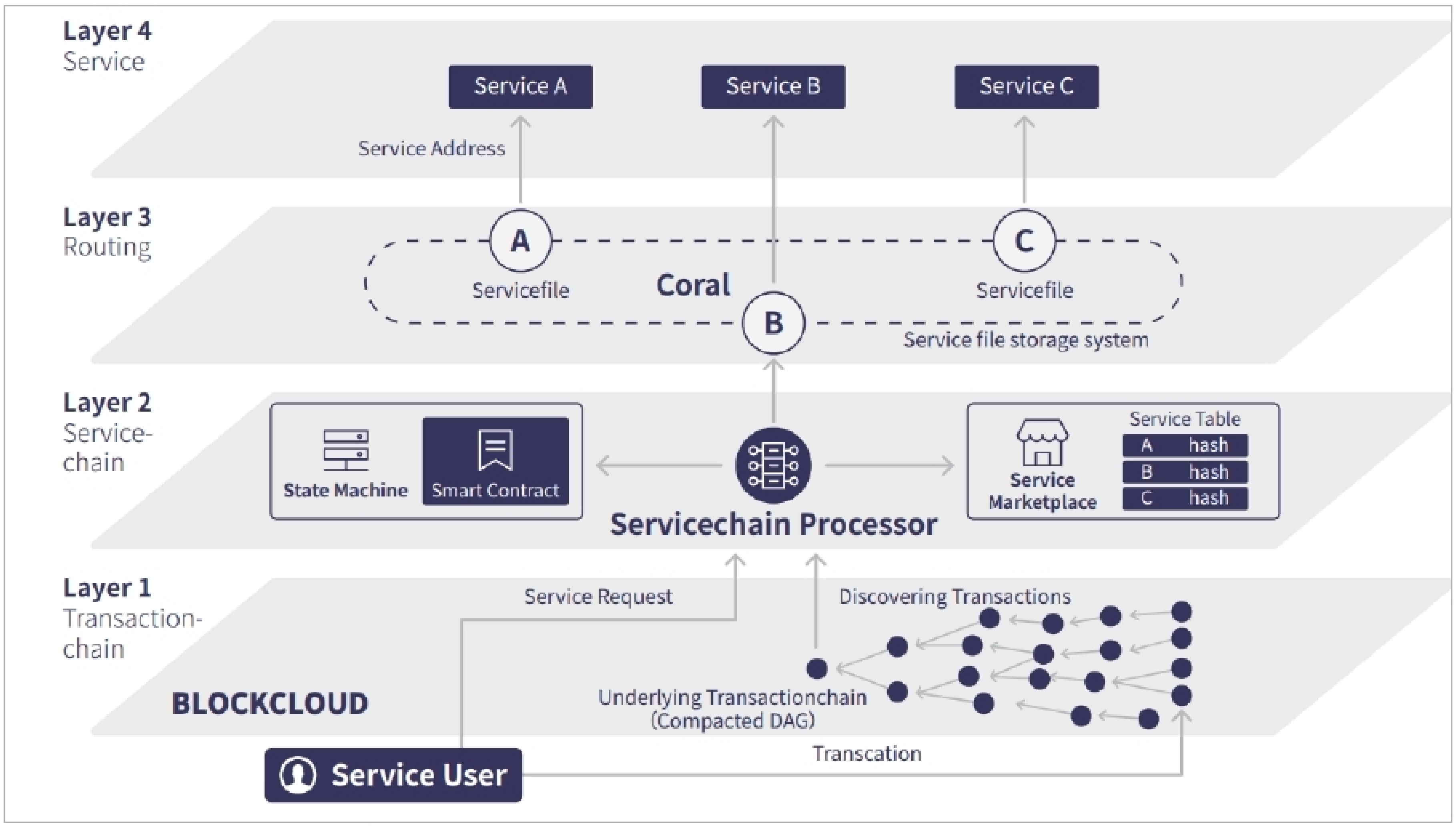

Figure 1: BlockCloud architecture (Source: *block-cloud.io*)

- *Sharing economy:* A decentralised, trustless, ecosystem powered by BlockCloud

BlockCloud architecture

BlockCloud provides new and enhanced functionalities on top of blockchains by providing new operations that are usually not supported by them. It has four layers — two under the control plane and two under the service plane. The transaction chain layer and service chain layer belong to the control plane, while the routing layer and service layer belong to the service plane.

Figure 1 highlights the BlockCloud architecture.

Transaction chain layer: This is a lower tier layer. All the operations are encoded in transactions on the underlying transaction chain. This layer stores all the network transactions as a global ledger. Due to IoT, the transaction chain layer should be elastic, scalable, cost-effective and secure. BlockCloud uses CoDAG for dynamic IoT networks, for best connectivity and low latency.

Service chain layer: The service chain layer defines new operations without making any significant changes to the blockchain. All BlockCloud operations are defined in the service chain layer and encoded in valid blockchain transactions as additional metadata. This layer gives the best support for processing BlockCloud operations. The rules for accepting or rejecting BlockCloud operations are also defined in the service chain. Accepted operations are processed by the service chain to construct a database that stores information on the global state of the system, along with state changes at any given blockchain block. Service chains can be used to build a variety of state machines. Currently, BlockCloud defines two state machines — a global service management system, and a service matching and pricing system.

Routing layer: The most significant characteristic of the blockchain is to separate the task of discovering routing requests from that of providing services. This reduces the need of the system to adopt any management service from the onset, and allows multiple service providers to exist -- both for commercial and peer-to-peer systems. Service chain binds names to the respective hash (service file) and stores these bindings in the control plane, whereas the service files themselves are stored in the routing layer. Users do not need to trust the routing layer because the integrity of service files can be verified by checking the hash (zone file) in the control plane.

Service layer: The topmost layer of BlockCloud is the service layer, the main aim of which is to provide services to the network. By providing services outside the transaction chain, BlockCloud allows arbitrary IoT services that are provided by a variety of IoT devices. Users do not need to trust the service provider because they can verify the authenticity of the service by proof of service in the control plane. At the core of BlockCloud is a new SAL that sits between the transport and network layers. The SAL maps service names in packets to network